

torch.diff#

<https://pytorch.org/docs/stable/generated/torch.diff.html>

Description:

Computes the n-th forward difference along the given dimension. The first-order differences are given by $\text{out}[i] = \text{input}[i + 1] - \text{input}[i]$. Higher-order differences are calculated by using `torch.diff()` recursively. `input` (Tensor) – the tensor to compute the differences on `n` (int, optional) – the number of times to recursively compute the difference `dim` (int, optional) – the dimension to compute the difference along. Default is the last dimension.

Function Signature

```
torch.diff(input, n=1, dim=-1, prepend=None, append=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.tensor([1, 3, 2])
>>> torch.diff(a)
tensor([ 2, -1])
>>> b = torch.tensor([4, 5])
>>> torch.diff(a, append=b)
tensor([ 2, -1,  2,  1])
>>> c = torch.tensor([[1, 2, 3], [3, 4, 5]])
>>> torch.diff(c, dim=0)
tensor([[2, 2, 2]])
>>> torch.diff(c, dim=1)
tensor([[1, 1],
        [1, 1]])
```

Detailed Documentation

Documentation

Computes the n -th forward difference along the given dimension.

The first-order differences are given by $\text{out}[i] = \text{input}[i + 1] - \text{input}[i]$. Higher-order differences are calculated by using `torch.diff()` recursively.

`input` (Tensor) – the tensor to compute the differences on

`n` (int, optional) – the number of times to recursively compute the difference

`dim` (int, optional) – the dimension to compute the difference along. Default is the last dimension.

`prepend` (Tensor, optional) – values to prepend or append to `input` along `dim` before computing the difference. Their dimensions must be equivalent to that of `input`, and their shapes must match `input`'s shape except on `dim`.

`append` (Tensor, optional) – values to prepend or append to `input` along `dim` before computing the difference. Their dimensions must be equivalent to that of `input`, and their shapes must match `input`'s shape except on `dim`.

`out` (Tensor, optional) – the output tensor.